Advanced Driver Assistance System

High-Level Design (HLD)

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This document describes the software architecture of the advanced driver assistance domain: camera processing, radar fusion, lane assistance and adaptive cruise control components, ports, interfaces, signals and flows.

1 Revision history

Revision	Date	Author	Description
C	2026-05-14	A. Kumar	Initial architecture baseline

2 Scope and context

2.1 Scope

The ADAS domain covers camera object detection, mid-range radar fusion, lane keeping assist and adaptive cruise control. Out of scope: raw sensor calibration, hardware wiring and production software details.

2.2 System context

The domain runs on AdasEcu with a dedicated vision accelerator. Gateway exchange with the vehicle motion domain is performed by AdasGateway over the chassis Ethernet link.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
CameraProcessingUnit	ApplicationSwComponentType	Camera object detection and classi
RadarFusionEngine	ApplicationSwComponentType	Mid-range radar target fusion
LaneAssistController	ApplicationSwComponentType	Lane keeping assist arbitration
AdaptiveCruiseController	ApplicationSwComponentType	Adaptive cruise control logic
ObstacleDetector	ApplicationSwComponentType	Static and dynamic obstacle tracki
AdasSignalService	ServiceSwComponentType	ADAS signal gateway service
AdasGateway	EcuAbstractionSwComponentType	Cross-domain gateway exchange
AdasFrontendComposition	CompositionSwComponentType	Composition of the ADAS components

3.2 Ports and interfaces

Port	Direction	Interface	Component
CameraObjectsOut	provided	ObjectListI	CameraProcessingUnit
RadarObjectsIn	required	RadarObjectListI	RadarFusionEngine
LaneOffsetOut	provided	LaneOffsetI	LaneAssistController
AccTargetOut	provided	AccTargetI	AdaptiveCruiseController
ObstacleListOut	provided	ObstacleListI	ObstacleDetector
AdasGatewayIn	required	AdasSignalSetI	AdasSignalService
GatewayToMotion	provided	MotionCommandI	AdasGateway
SteeringCommandOut	provided	SteeringCommandI	LaneAssistController

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
CameraObjectList	ObjectListI	CameraProcessingUnit	RadarFusionEngine	64
RadarObjectList	RadarObjectListI	RadarFusionEngine	ObstacleDetector	48
LaneOffset	LaneOffsetI	LaneAssistController	AdasGateway	12
AccTargetDistance	AccTargetI	AdaptiveCruiseControlle	GatewayToMotion	16
ObstacleClass	ObstacleListI	ObstacleDetector	CameraProcessingUnit	8
SteeringCommand	SteeringCommandI	LaneAssistController	GatewayToMotion	16
AdasMode	AdasModeI	AdaptiveCruiseControlle	LaneAssistController	4

4 Functional flows

4.1 Camera object flow

CameraProcessingUnit detects objects and publishes CameraObjectsOut over ObjectListI. RadarFusionEngine consumes CameraObjectsOut together with RadarObjectsIn and produces a fused ObstacleListI consumed by ObstacleDetector for downstream arbitration.

4.2 Lane keeping flow

LaneAssistController computes LaneOffset from camera lane markers and requests a steering correction through SteeringCommandI. AdaptiveCruiseController consumes AdasMode to suppress lane requests during manual override.

4.3 Adaptive cruise flow

AdaptiveCruiseController tracks AccTargetDistance from the fused radar list and requests torque modulation from PowertrainGateway through MotionCommandI via AdasGateway.

5 Timing and runnables

TimingEvent CameraFrameEvent triggers ProcessCameraFrameRunnable every 40 ms. TimingEvent RadarSweepEvent triggers FuseRadarTargetsRunnable every 20 ms. TimingEvent LaneUpdateEvent triggers UpdateLaneAssistRunnable every 25 ms. All runnables are scheduled by AdasTask on AdasEcu.

6 Basic software integration

Signal routing uses Com and PduR. Incoming Ethernet frames are handled by SoAd and TcpIp. NvM stores the camera calibration block. Rte provides the port API used by all ADAS application components.

Mode management is performed by BswM and EcuM. Watchdog supervision for AdasTask is performed by WdgM. The vision accelerator interface uses Xcp for online calibration exchange.

The diagnostic path routes UDS requests through Dcm and Dem. AdasSignalService consumes AdasSignalSetI and forwards fault reports to the tester over DoIP.

7 Diagnostics and configuration

DTCs for camera blockage, radar misalignment and fusion timeout are defined in the diagnostics specification. The HLD only defines the reporting path: component to Dcm to SoAd.

Configuration parameters are grouped into the domain configuration container and written through NvM during shutdown.

8 Known limitations

CameraProcessingUnit degrades gracefully under heavy rain, but full performance recovery after blockage is not specified in this revision.

9 Open questions

Open: calibration ownership for cross-domain interfaces and the timeline for production software freeze are tracked outside this document.